Yun Chen TSAI

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SUMMARY

I received my Bachelor in mathematics from Hong Kong University of Science and Technology (HKUST) in 2024. I am currently a ph.D. student in informatics at National Institute of Informatics (NII), Japan, working under the supervision of Prof. Ichiro Hasuo. My research interest mainly falls in the large area of formal methods and model checking, with a particular focus on probabilistic and quantum model checking. Before my Ph.D., I also worked with Prof. Shing Yu Leung and Prof. Amir Goharshady at Hong Kong University of Science and Technology (HKUST) on projects related to dynamic systems, parameterized algorithms and program analysis.

EDUCATION

Hong Kong University of Science and Technology (HKUST)
BSc in Mathematics *Graduation: January 2024*

PUBLICATION

- Y. C. Tsai, K. Phalakarn, S. Akshay, I. Hasuo (2025). *Chance and Mass Interpretations of Probabilities in Markov Decision Processes.* To be appeared in International Conference on Concurrency Theory, CONCUR 2025.
- JF Raskin, Y. C. Tsai (2024). *Synthesis from LTL with Reward Optimization in Sampled Oblivious Environments.* Dependable Software Engineering. Theories, Tools, and Applications, SETTA 2024
- Y. C. Tsai, S. Y. Leung (2024). *Local Trajectory Variation Exponent (LTVE) for Visualizing Dynamical Systems.* Communication in Computational Physics, CiCP.
- G. K. Conrado, A. K. Goharshady, K. Kochekov, Y. C. Tsai, A. K. Zaher (2023). *Exploiting the Sparseness of Control-flow and Call Graphs for Efficient and On-demand Algebraic Program Analysis.* ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications, OOPSLA 2023.

Professional Services

- Artifact Evaluation Committee Member, TACAS 2025.

Awards and Scholarship

- Japan-Taiwan Exchange Association Scholarship
- Chern Class Talent Scholarship
- HKSAR Government Scholarship Fund - Reaching Out Award